

Casimir Repulsion

Attempting Stability

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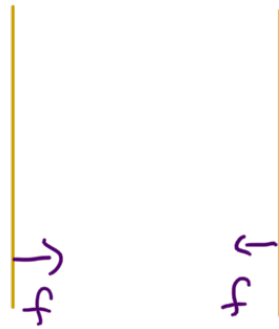
Thanks: Avinash Khatri Christian Rose

Southern Illinois University - Carbondale

Summer 2020

Outline

1.) What is Casimir Effect



Casimir 1948

2.) Casimir Repulsion



Boyer 1973

Boyer 1974



Levin 2010

Levin, McCarthy, Rodriguez, Reid Johnson 2010

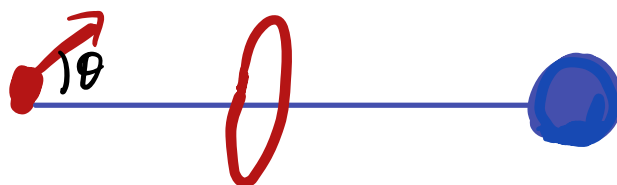
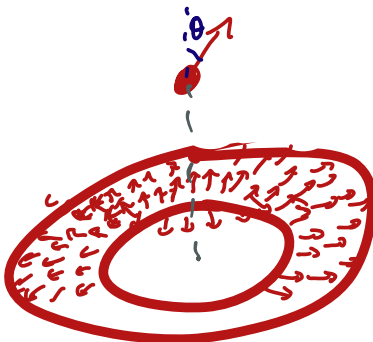


Shajesh Schaden 2012

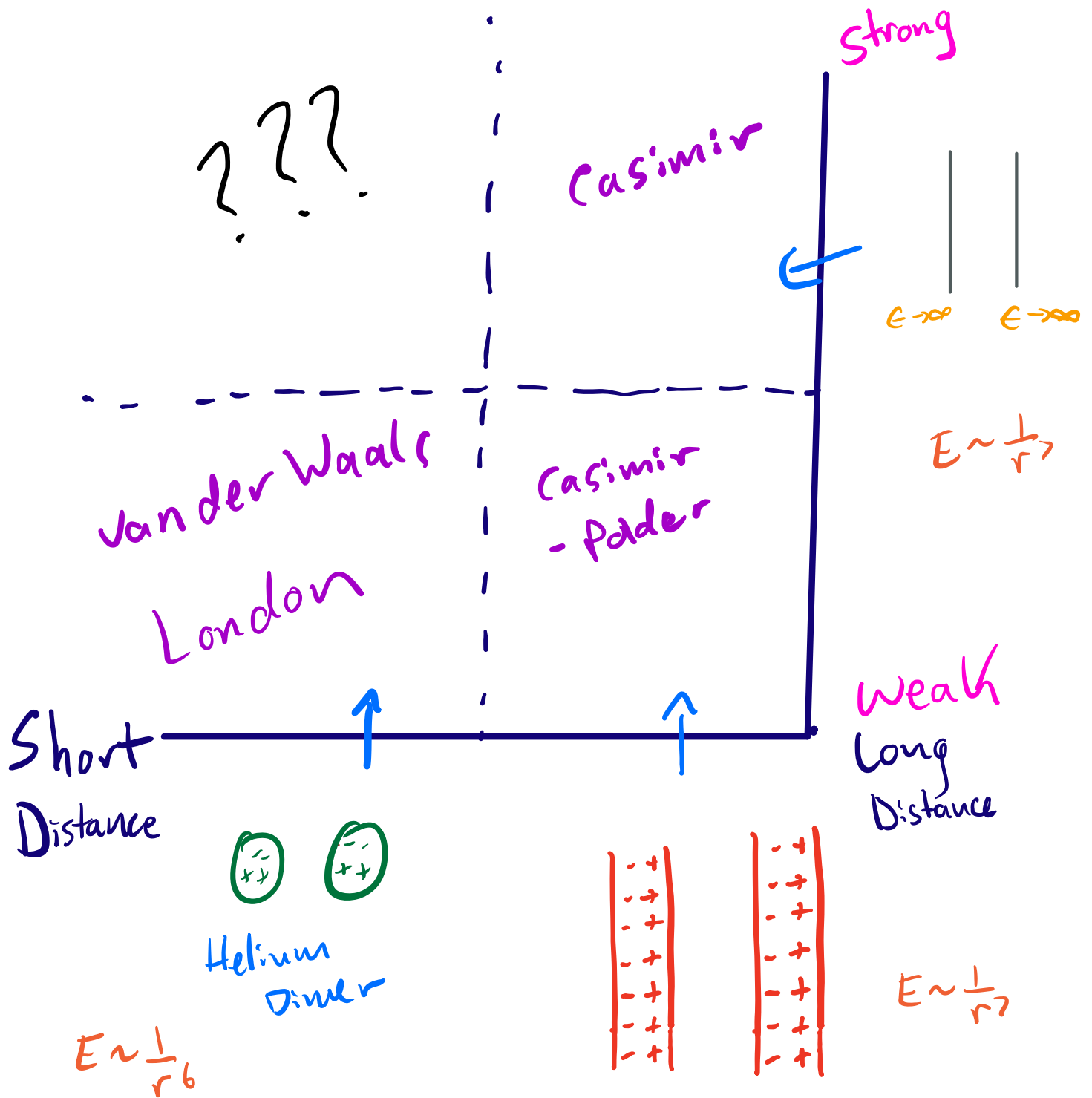
Shajesh Schaden 2012

3.) New findings

Marchetta Parashar 2020
Shajesh



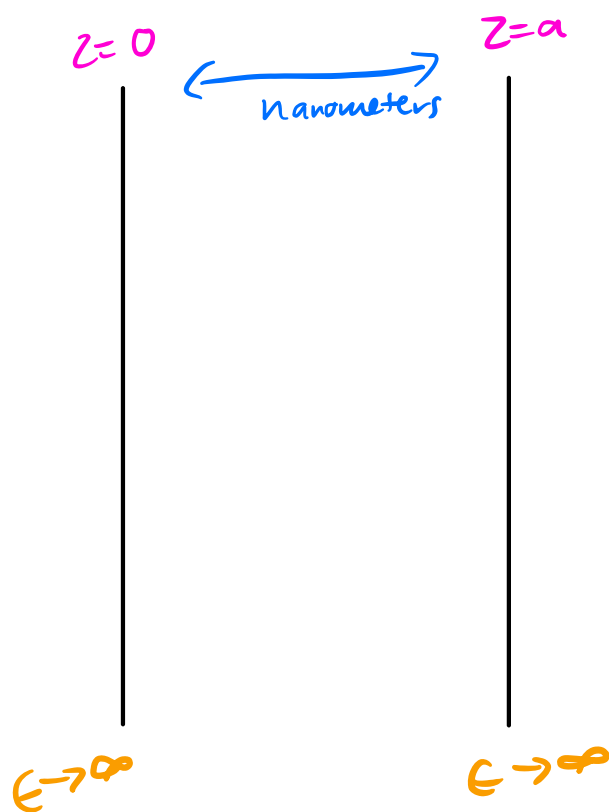
Quantum E&M Forces



Casimir Effect

One of the least intuitive concepts
in Quantum Mechanics.

Attraction between neutral objects
in the Vacuum (Empty space)



"Macroscopic bodies
effect the field,

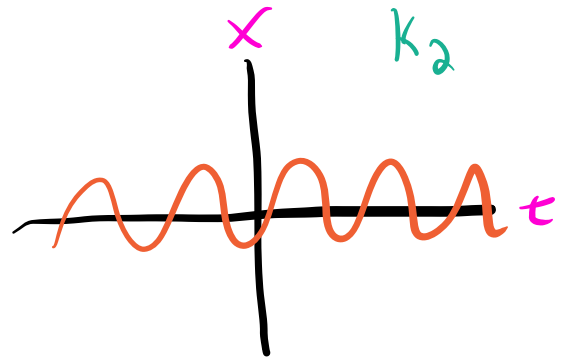
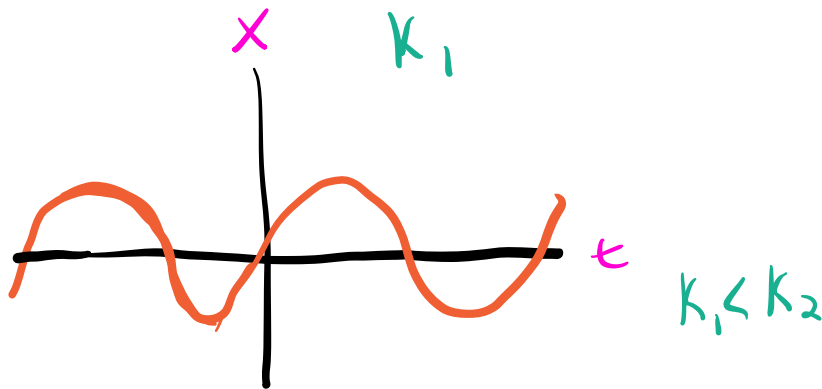
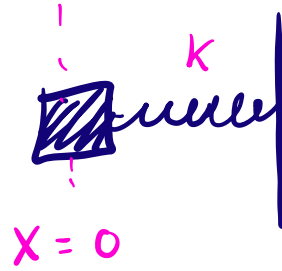
the field
acts on the bodies."

$$\langle E \rangle = 0$$

$$\langle E E \rangle \neq 0$$

Zero Point Energy

Classical
Harmonic Motion:

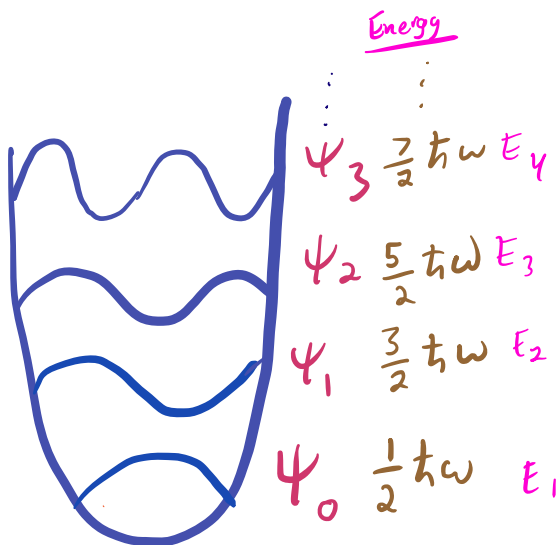


Quantum Mechanics:

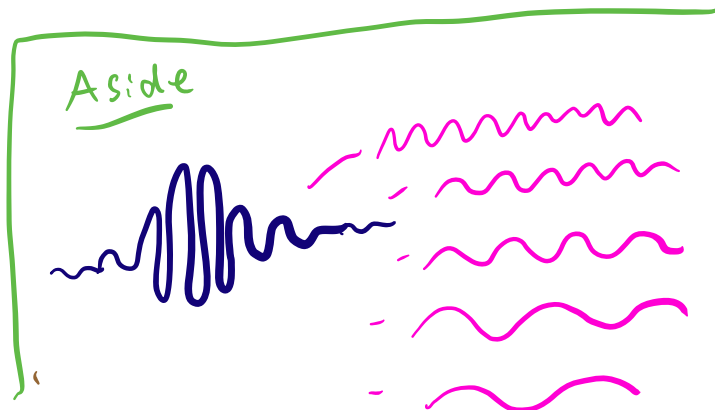
$$\Delta x \Delta p \geq \frac{\hbar}{2} \sim 10^{-34}$$

Particle $\Rightarrow$ Probability distribution

$$H \psi = E \psi$$



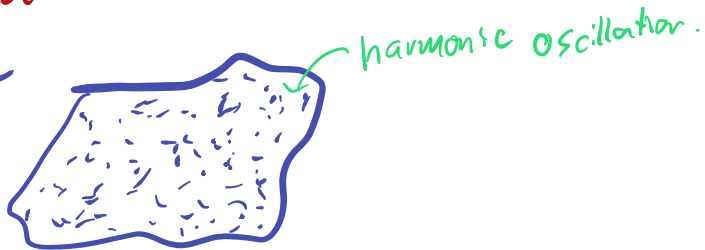
$$E_n^{(\text{Particle})} = \left(n + \frac{1}{2}\right) \hbar \omega$$



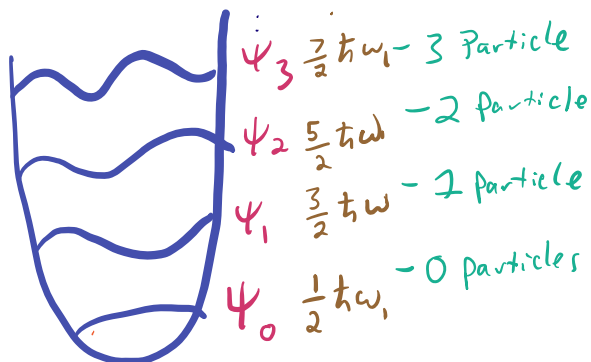
Zero Point Energy

Quantum Field $\Rightarrow$ ∞ many harmonic oscillators

universe

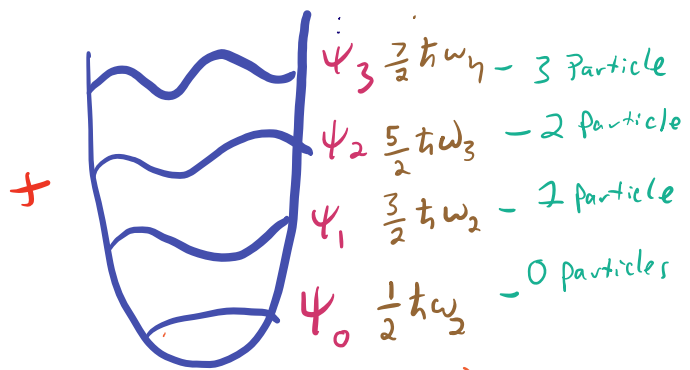


Particles w/ momentum ω_1



$$E_n^1 = (n + \frac{1}{2}) \hbar \omega_1$$

Particles w/ momentum ω_2 - - - -



$$E_n^2 = (n + \frac{1}{2}) \hbar \omega_2$$

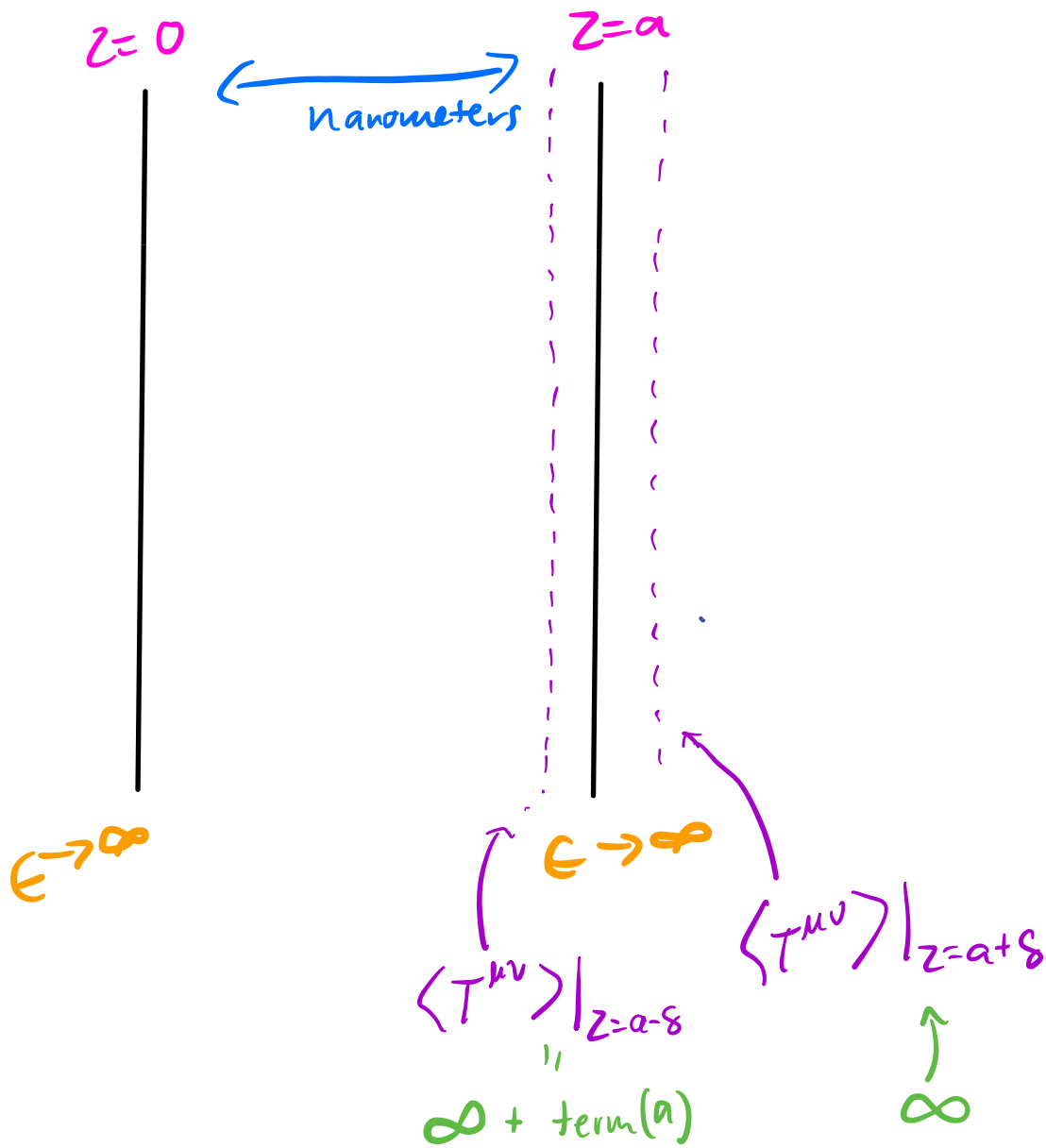
$$E^{(\text{field})} = \sum_{k=0}^{\infty} E_n^k = \sum_{k=0}^{\infty} (n + \frac{1}{2}) \hbar \omega_k$$

Vacuum state = No particles

$$E = \sum_{k=0}^{\infty} (0 + \frac{1}{2}) \hbar \omega_k = \frac{1}{2} \sum_{k=0}^{\infty} \hbar \omega_k = \underline{\infty}$$

Zero point energy of quantum field

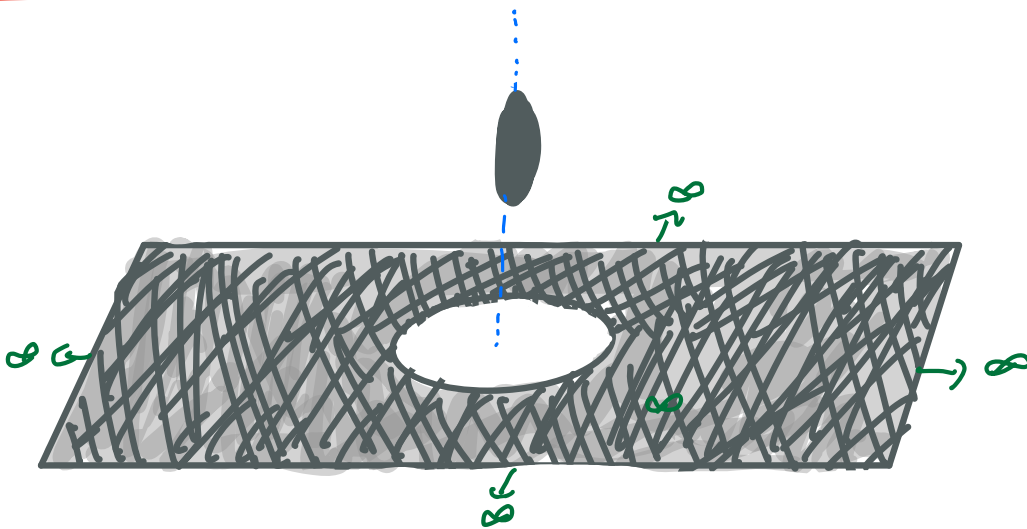
Returning to Casimir Effect



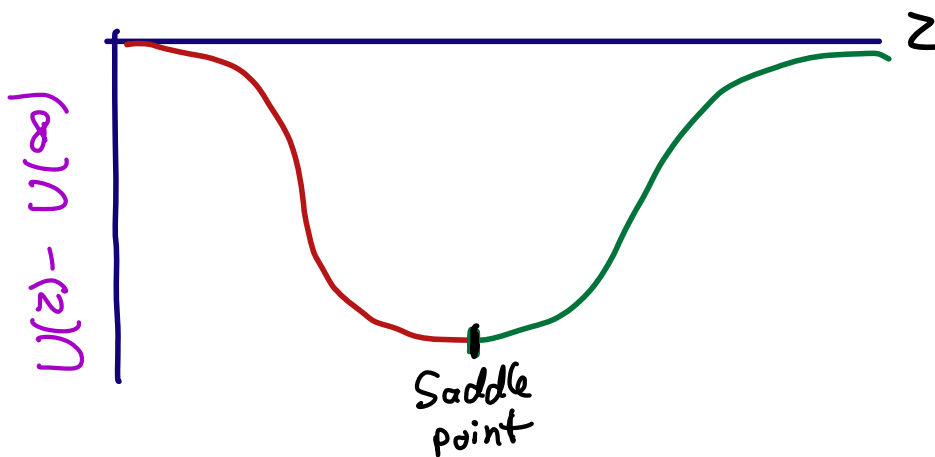
$$P_{\text{pressure}} = \frac{F \cdot \hat{z}}{A} = \langle T^{\mu\nu} \rangle|_{z=a} - \langle T^{\mu\nu} \rangle|_{z=a+\delta}$$

$$= \infty + \text{term}(a) - \infty = -\frac{\pi^2 \hbar c}{240 a^4}$$

Anisotropic Casimir Repulsion



- Calculated Numerically
- Interaction Energy $U(0) = U(\infty) = 0$

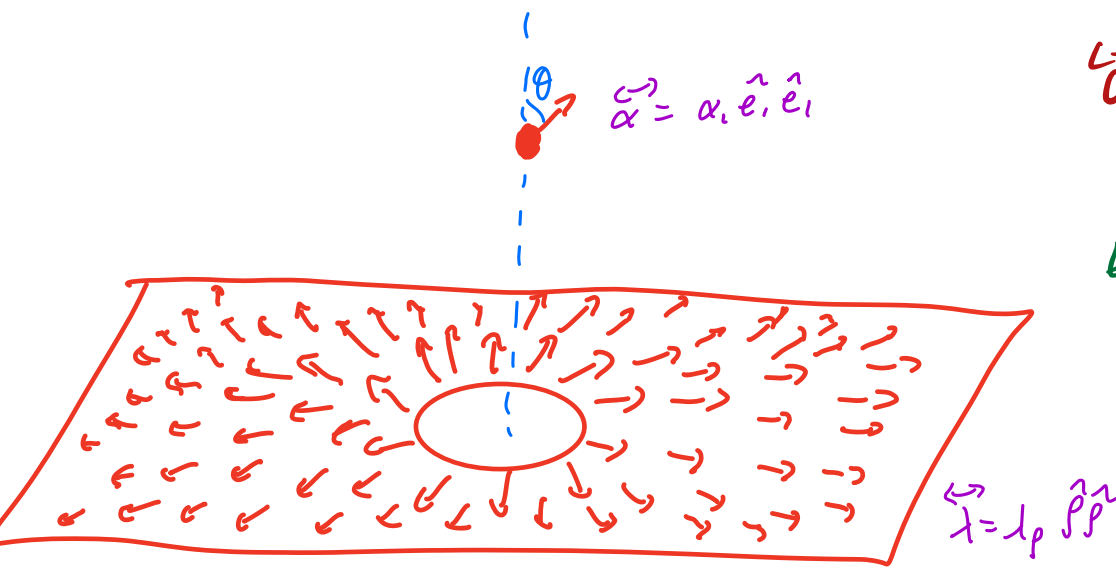


Unstable! Lateral Perturbations.

(Rahi, Kardar, Emig 2007)

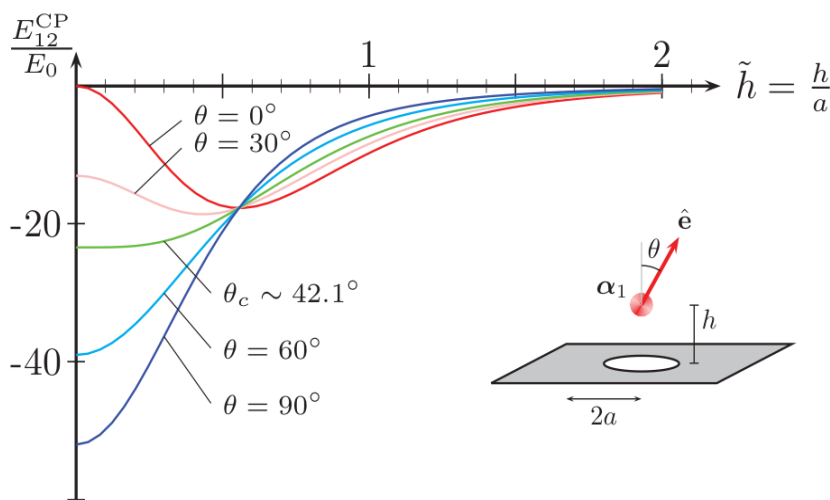
Obeys a generalization of Earnshaw's theorem concluding neutral objects cannot be in static equilibrium under thermal / quantum fluctuations.

Anisotropic Casimir Repulsion pt. 2



$$\vec{\alpha} = \begin{pmatrix} \alpha_1 & 0 & 0 \\ 0 & \alpha_2 & 0 \\ 0 & 0 & \alpha_3 \end{pmatrix}$$

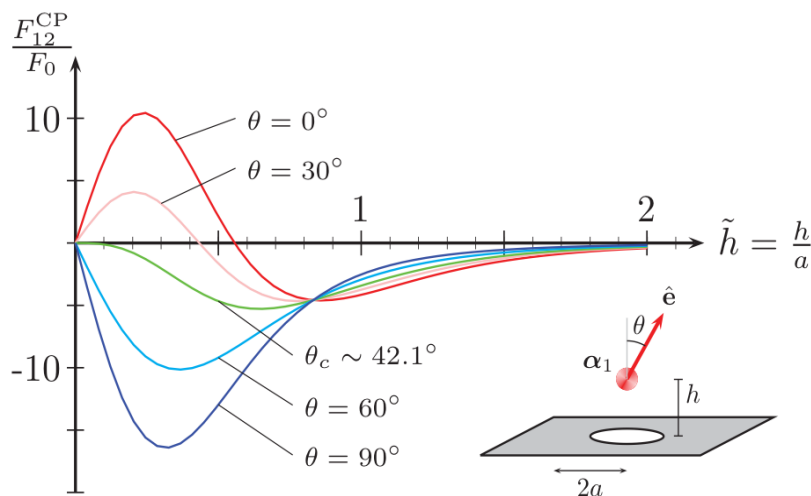
$$\vec{\lambda} = \begin{pmatrix} \lambda_s & 0 & 0 \\ 0 & \lambda_\phi & 0 \\ 0 & 0 & \lambda_z \end{pmatrix}$$



repulsive
only if
 $\alpha_1 \neq \alpha_2 \neq \alpha_3$

$$\lambda_s \neq \lambda_z$$

$$\lambda_\phi \neq \lambda_z$$



Having
 $\lambda_s = \lambda_\phi$
is okay!

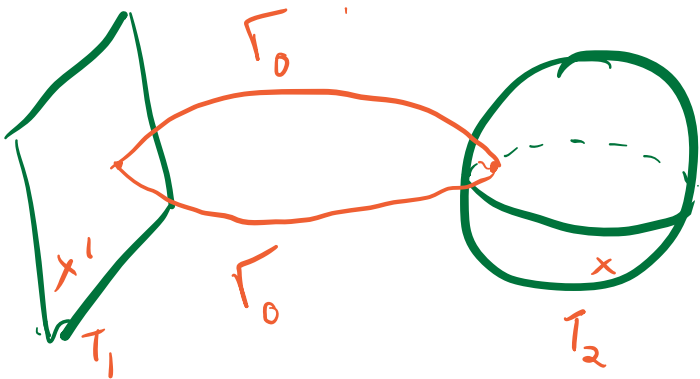
This calculation was made possible by
Weak Approximation $E - 1 \ll 1$

Formalism: The ∞ energy we saw can be divided into

$$E_{\text{total}} = \underbrace{E_0}_{\substack{\text{vacuum} \\ \downarrow \\ \infty}} + \underbrace{E_1}_{\substack{\text{object 1} \\ \downarrow \\ \infty}} + \underbrace{E_2}_{\substack{\text{object 2} \\ \downarrow \\ \infty}} + \underbrace{E_{12}}_{\substack{\text{Interaction} \\ \downarrow \\ \text{finite}}}$$

Extracting the finite term

$$E_{12} = \frac{\hbar c}{32\pi^2} \int_{-\infty}^{\infty} d\omega \int_{-\infty}^{\infty} d^3x \int_{-\infty}^{\infty} d^3x' \underbrace{\text{Tr} \ln [1 - \Gamma_0(x, x'; \omega) T_1(x) \Gamma_0(x, x'; \omega) T_2(x')] - \text{Tr} [\Gamma_0(x, x'; \omega) T_1(x) \Gamma_0(x, x'; \omega) T_2(x')]}_{\text{finite}}$$



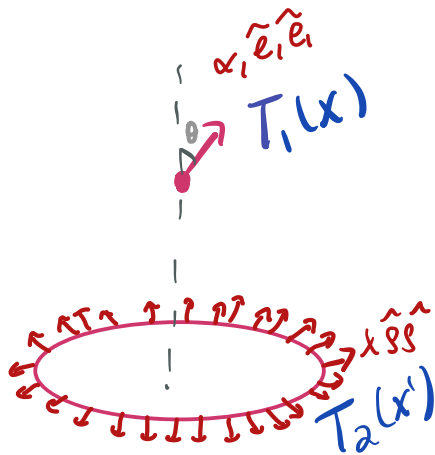
T_1, T_2 encoding the geometry and polarization of the material

$\Gamma_0(x, x'; \omega)$ Propagates a virtual particle from x, x' satisfying

$$[\nabla_x \nabla_{x'} - \vec{1}] \Gamma_0(x, x'; \omega) = \delta(x - x')$$

$$\Gamma_0 \sim \frac{e^{-|\omega|r}}{r^3}$$

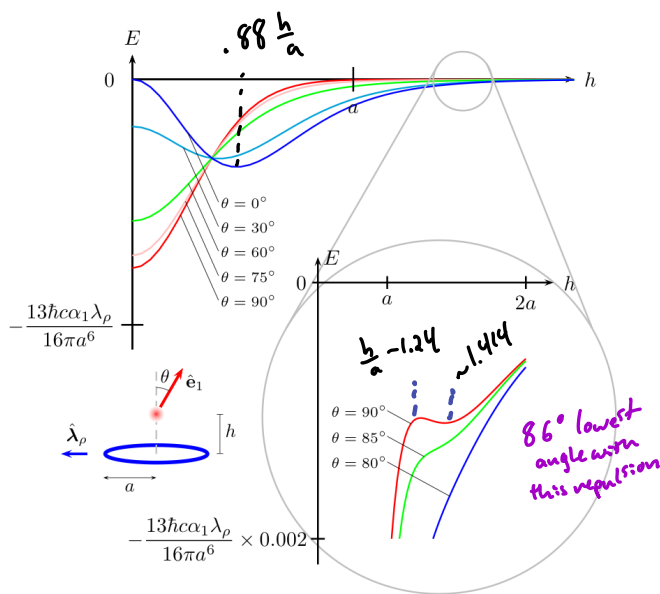
Ring-Atom Example



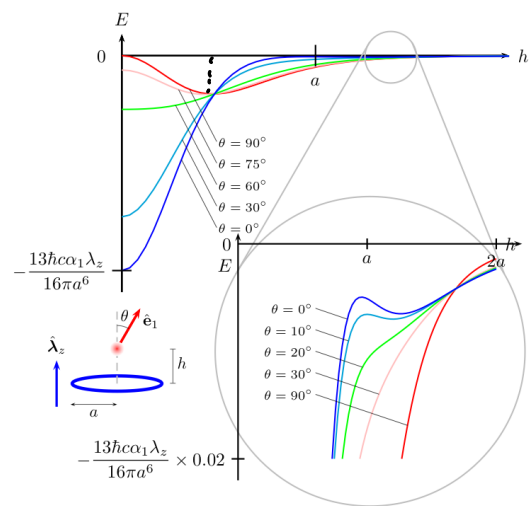
$$T_1(x) \sim \alpha_i \hat{e}_i \hat{e}_i \delta^3(\vec{x} - \vec{x}')$$

$$T_2(x) \sim \lambda \hat{s} \hat{s} \delta(\rho - a) \delta(z)$$

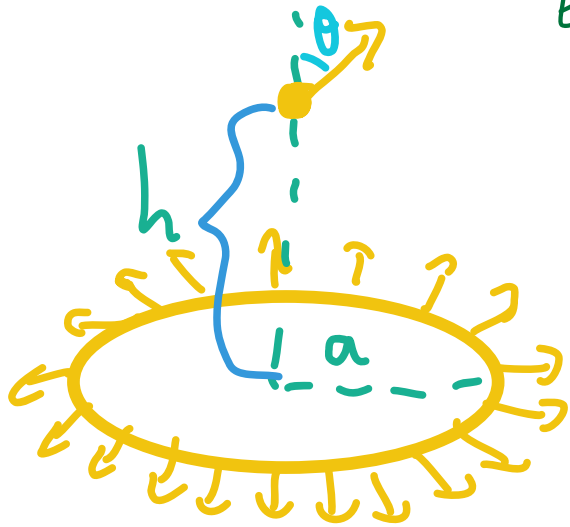
$$E_{12} = \frac{-hc}{32\pi^2} \int_0^{2\pi} d\phi \frac{1}{r^2} \left[13 (\hat{e}_i \cdot \hat{s})^2 - 56 (\vec{r} \cdot \hat{e}_i) (\vec{r} \cdot \hat{s}) + 63 (\vec{r} \cdot \hat{e}_i)^2 (\vec{r} \cdot \hat{s})^2 \right]$$



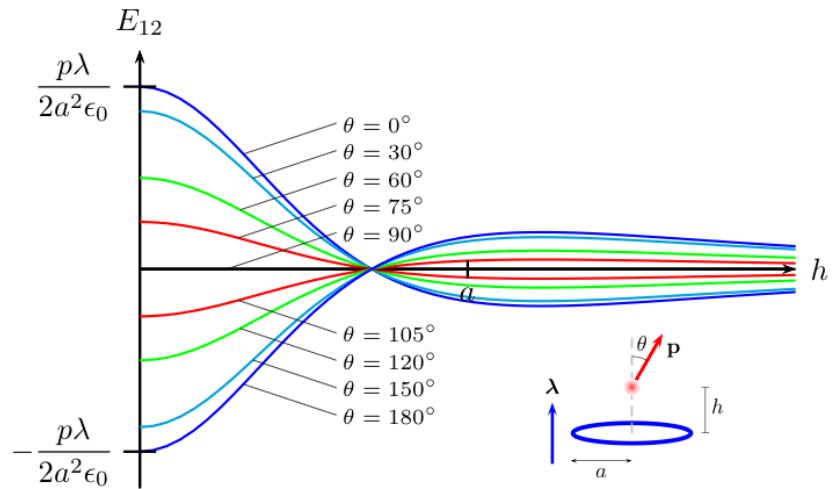
Similarly



Compared to classical Permanent EM Dipole



$$E_{12} = \frac{1}{4\pi\epsilon_0} \int_0^{2\pi} d\phi' \frac{[(\mathbf{p} \cdot \mathbf{r}) - 3(\mathbf{p} \cdot \hat{\mathbf{r}})(\hat{\mathbf{r}} \cdot \mathbf{r})]}{r^3}$$



	$\theta = 0$	$\theta = 90$	$\theta = 180$
Quantum induced dipole			
Permanent dipole		no force	

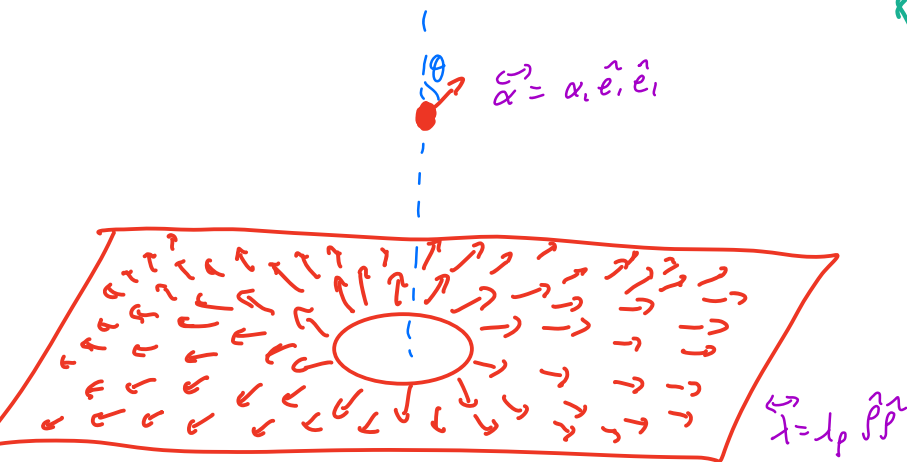
We assumed anisotropy for the
atom + ring : $\alpha_1 \neq \alpha_2 \neq \alpha_3$ $\lambda_g \neq \lambda_\phi \neq \lambda_z$

But we saw in the plate that relaxing $\lambda_g = \lambda_\phi$
gives us the same qualitative features.

Both anisotropy and $\lambda_g = \lambda_\phi$

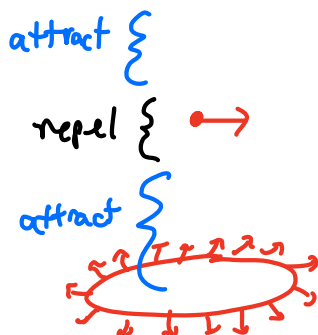
For $\theta < 42^\circ$ Short distance
repulsion
Long distance
attraction

For $\theta > 42^\circ$ All attraction

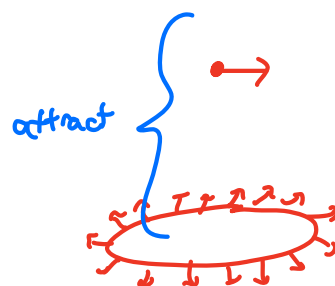


However, the New repulsive region $\theta > 85^\circ$
Vanishes for Planar isotropy

$$\lambda_g \neq \lambda_\phi \neq \lambda_z$$

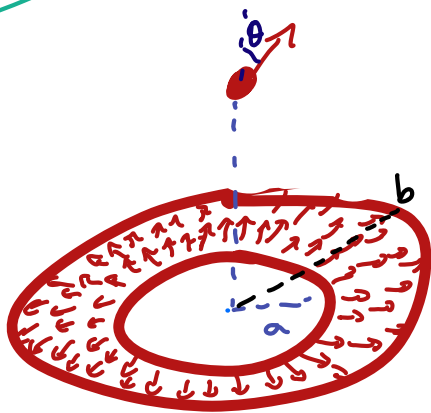


$$\lambda_g = \lambda_\phi$$



The intermediate repulsive region for $\theta > 85^\circ$
 must appear when contracting a plate with
 a hole to a ring

Disk + Atom



$$T_{\text{atom}} = \vec{\nabla} \delta^3(x-x_0)$$

$$T_{\text{Disk}} = \vec{\nabla} [\theta(b-\rho) \theta(\rho-a)] \delta(z)$$

$$\vec{\nabla} = \alpha_i \hat{e}_i, \quad \vec{\nabla} = \sigma_\rho \hat{\rho} \hat{\phi}$$

$$\sigma_\rho = \frac{1}{(b-a)}$$

Aside $b \rightarrow \infty$ gives us plate + atom

$$\lim_{b \rightarrow \infty} \vec{\nabla} [\theta(b-\rho) \theta(\rho-a)] = \lim_{b \rightarrow \infty} \left[\frac{1}{(b-a)} [\theta(b-\rho) \theta(\rho-a)] \right] = 1 \delta(\rho-a) \Rightarrow \text{Ring + Atom}$$

repulsion ceases

$\theta = 90^\circ$	$b = 1.25 a$
$\theta = 88.2^\circ$	$b = 1.2 a$
$\theta = 87.3^\circ$	$b = 1.1 a$

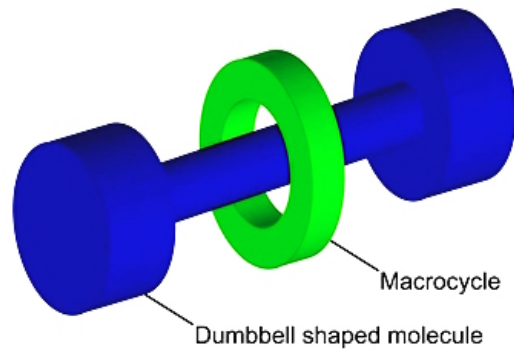
For $\theta \leq 87^\circ$ No repulsion exists

Casimir Engine

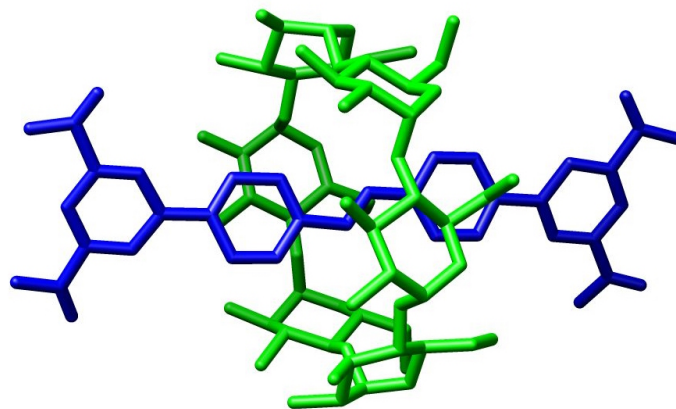
Casimir polder approx. requires $\sim 100\text{nm}$

Rotaxane

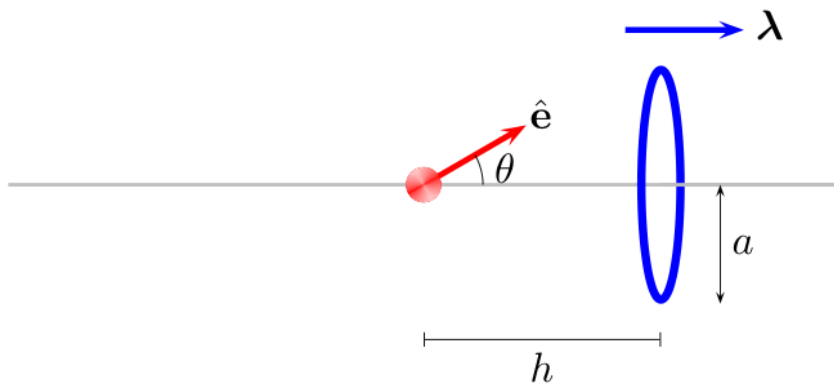
Ideal:



Reality:

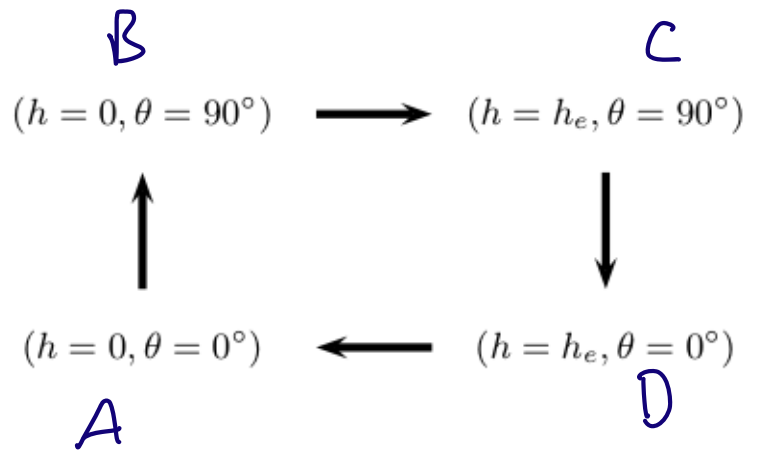
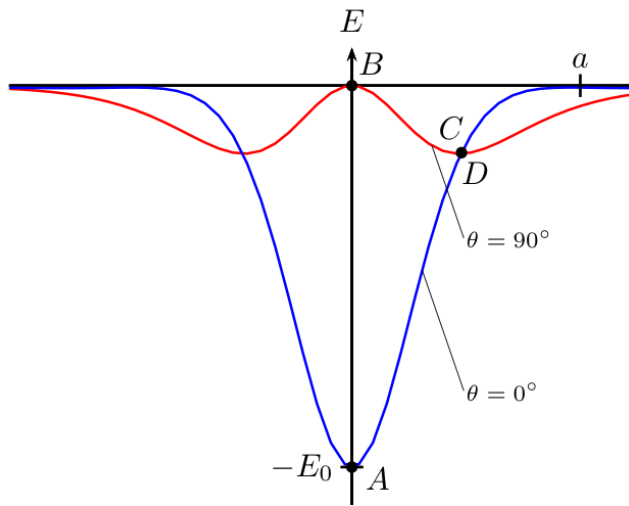


We saw



For $\theta=0$, the ring is attracted

For $\theta=90^\circ$, the ring is repelled



$$W_{A \rightarrow B} = - \int_0^{\pi/2} \tau d\theta = E(0, \frac{\pi}{2}) - E(0, 0) > 0$$

$$W_{B \rightarrow C} = - \int_0^{h_e} F dh = E(h_e, \frac{\pi}{2}) - E(0, \frac{\pi}{2}) < 0$$

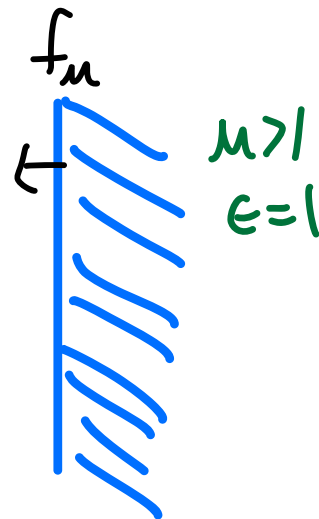
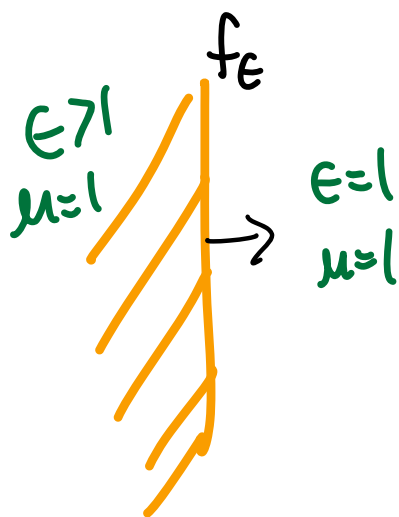
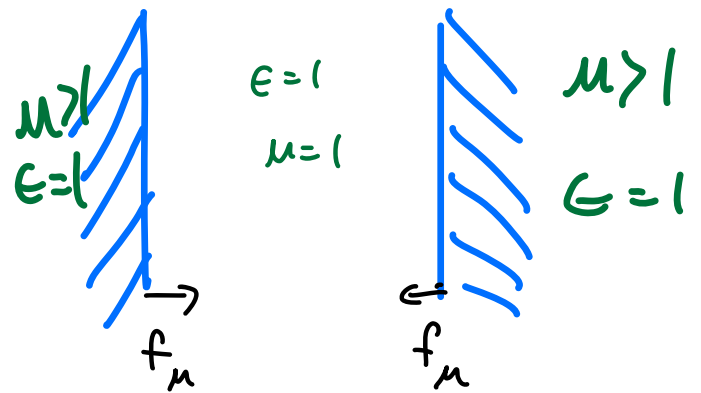
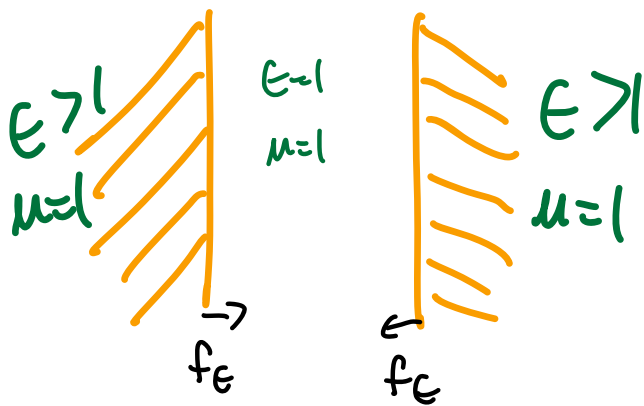
$$W_{C \rightarrow D} = - \int_{\pi/2}^0 \tau d\theta = E(h_e, 0) - E(h_e, \frac{\pi}{2}) = 0$$

$$W_{D \rightarrow A} = - \int_{h_e}^0 F dh = E(0, 0) - E(h_e, 0) < 0$$

Classical Electrodynamics

$$f_E = -\frac{1}{2} E^2 \nabla \epsilon$$

$$f_M = -\frac{1}{2} B^2 \nabla \mu$$



Using Zero Point Energy Method's

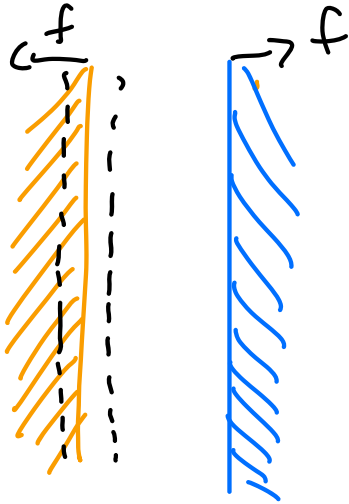
They Repel



The Van der Waal's interaction for any
purely electric and purely magnetic atoms will
repel



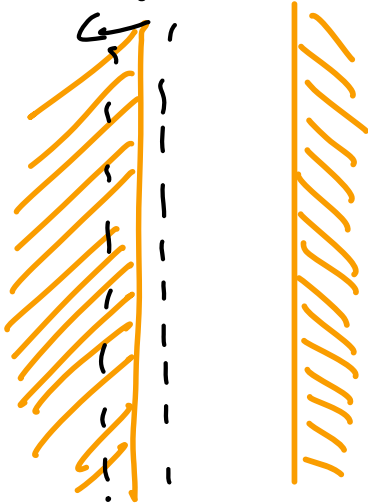
Electric - Magnetic repulsion.



$$P = -\frac{1}{\pi^2} \int_0^{\infty} \frac{k^3 dk}{e^{2ka} + 1} = -\frac{7}{8} \frac{\pi^2}{240} a^4$$

↑
Fermi Dirac
distribution

$$P = \langle T^{\mu\nu} \rangle_- - \langle T^{\mu\nu} \rangle_+$$



$$P = \frac{1}{\pi^2} \int_0^{\infty} \frac{k^3 dk}{e^{2ka} - 1} = \frac{\pi^2}{240} a^4$$

↑
Bose-Einstein
distribution

$$\frac{7}{8} = \frac{\int_0^{\infty} \frac{k^3 dk}{e^{2ka} + 1}}{\int_0^{\infty} \frac{k^3 dk}{e^{2ka} - 1}}$$

Spinning the atom



$$\tau = - \frac{\partial E_{12}}{\partial \theta}$$

$$= \frac{-\alpha \hbar \sin 2\theta}{32\pi^2} \frac{a(26a^4 - 123\hbar^2 a^2 + 40\hbar^4)}{(a^2 + \hbar^2)^{1/2}}$$

How to find

$$\frac{dL}{dt} = r \times \vec{F}$$

Questions to answer, What vacuum fluctuation induced angular momentum mean?

$$K = \vec{\nabla} \times \langle T^{\mu\nu} \rangle = \frac{1}{i\hbar} \lim_{r \rightarrow r'} \vec{\nabla} \times \overleftrightarrow{\Gamma}(r, r') \quad \overleftrightarrow{\Gamma} = \frac{\overleftrightarrow{\Gamma}_0}{1 - \chi \overleftrightarrow{\Gamma}_0}$$

Two Neutral Atoms



Classically, it costs more energy to create a dipole than what one gets back from fluctuations. $\langle d \rangle = 0$ $\langle d^2 \rangle = 0$

QM: Zero Point Energy of the oscillator of the atomic dipole moment gives $\langle d \rangle = 0$ $\langle d^2 \rangle \neq 0$

Say atom A spontaneously fluctuates into a dipole

$$E_{\text{dipole A}} \propto \frac{d_A}{R^3}$$

Which stimulates $d_B \propto E_{\text{dip A}} \sim \frac{d_A}{R^3}$

Now, the expression for dipole-dipole interaction:

$$E_{\text{int}} = \frac{1}{2\pi\epsilon_0} \frac{d_A d_B}{R^3} = \frac{1}{2\pi\epsilon_0} \frac{\langle d_A^2 \rangle}{R^6}$$

This is the Van-der Waal's interaction

When the separation distance is greater than the wavelengths of the atoms
We get Casimir Effect which goes $\sim \frac{1}{R^7}$

